

Calculus I Lesson 24

- 1) If a pebble drops in water, creating a circular ripple where the radius expands at a rate of $\frac{dr}{dt}=12$ inches per second, what is the rate of change of the circumference?

Circumference of a circle: $C = 2\pi r$

- a) Differentiate both sides of the equation $C = 2\pi r$ with respect to t (time).

Implicit differential equation is _____?

- b) Find $\frac{dC}{dt} = \underline{\hspace{1cm}}?$, given $\frac{dr}{dt}=12$.

Note: $\frac{dC}{dt}$ is rate of change circumference with respect to t (time)

2) The area of a circle increases at a rate of $10 \text{ cm}^2/\text{s}$

How fast is the radius changing when the radius is 2cm ?

Area of circle: $A = \pi r^2$

a) Differentiate both sides of the equation $A = \pi r^2$ with respect to t (time).

Implicit differential equation is _____?

b) Find $\frac{dr}{dt} = \underline{\hspace{1cm}}?$, given $\frac{dA}{dt} = 10$.

Note: $\frac{dr}{dt}$ is rate of change radius with respect to t (time)

3) Air is pumped into a spherical balloon at a rate of $350 \text{ cm}^3/\text{min}$.

How fast is the radius expanding when the radius is 10 cm?

Volume of sphere: $V = \frac{4}{3}\pi r^3$

a) Differentiate both sides of the equation $V = \frac{4}{3}\pi r^3$ with respect to t (time).

Implicit differential equation is _____?

b) Find $\frac{dr}{dt} = \underline{\hspace{2cm}}?$, given $\frac{dV}{dt} = 350$.

Note: $\frac{dr}{dt}$ is rate of change radius with respect to t (time)

- 4) A cylindrical tank with radius 5 meters is being filled with water at a rate of $4 \text{ m}^3 / \text{min}$.
How fast is the height of the water increasing?

Volume of cylinder: $V = \pi r^2 \cdot h$

Note: $V = \pi r^2 \cdot h = V = \pi(5)^2 \cdot h = 25\pi h$.

- a) Differentiate both sides of the equation $V = 25\pi h$ with respect to t (time).

Implicit differential equation is _____?

- b) Find $\frac{dh}{dt} = \underline{\hspace{2cm}}$, given $\frac{dV}{dt} = 4$ and .

